MID SEM 1: MTH201

Name: [redacted] Reg No.: [redacted] Time: 1 h Marks: 20

Q 1. Check if the following limit exists. If it exists, then calculate the limit.

(i) $\lim_{x\to 1} \dfrac{|x-1|}{(x-1)x}$ [3]

(ii) $\lim_{x\to 0} x \cos\left(\dfrac{1}{x}\right)$ [2]

Q 2. (i) Let $f(x) = \begin{cases} 2x & \text{if } x \geq 1 \\ x & \text{if } x < 1. \end{cases}$

Show that the function is discontinuous at $x = 1$.
Show in terms of ϵ and δ that f is discontinuous at $x = 1$. [3]

(ii) Show that $x^3 + x^2e^{-x} - 5 = 10$ has atleast one solution in the interval $[0, 4]$ [2]

Q 3. (i) Show that $||x| - |y|| = |x - y|$ if and only if $|xy| = xy$. [2.5]

(ii) Let $a \in \mathbb{R}$, such that $a \leq 1 + \epsilon$ for every $\epsilon > 0$. Show that $a \leq 1$. [2.5]

Q 4. (i) Calculate the derivative y' if $x^2y^4 = 3$. [2]

(ii) Show that the extremum points of the function $h(x) = (x^2 - 1)e^{2-x^2}$ are $0, \pm\sqrt{2}$. Show, from the definition of local, that 0 is a point of local minimum. [3]